



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

UNIT TEST

24-07-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A	C	B,C	A,B,C,D	A,C	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A,B,C	B,D	B	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	9.00	5.00	5.00	10.00	5.00	10.00	3.00	2.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C,D	A,C,D	A,B,C,D	A,C,D	A,C,D	A,B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	A	A,B,C	A,B,D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	6.00	4.00	6.00	6.00	5.00	3.00	4.00	6.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,D	A,B,C	A,B,C,D	A,B,C	A,C,D	A,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B,D	A,D	A,B,D	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.25	6.50	0.94 to 0.96	8.00	1.00	0.00	2.00	0.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A)

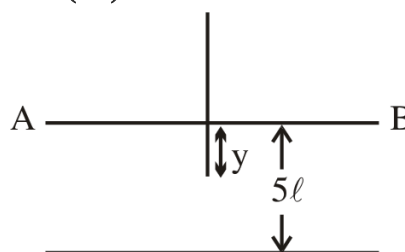
In com frame both the masses execute S.H.M.

$$w = \sqrt{\frac{K}{m}} = \sqrt{\frac{2K}{m}}$$

Initial particles are at extreme position distance

$$= L_0 + (L - L_0) \cos \sqrt{\frac{2K}{m}} t$$

2. Ans (C)



$$a = \frac{\left(\frac{m}{\ell}\right) yg}{m} = \frac{gy}{\ell}$$

$$\int_0^u v dv = \frac{g}{\ell} \int_0^{\ell} y dy$$

$$u = \sqrt{g\ell} \quad \dots(1)$$

$$v^2 = u^2 + 2as = (\sqrt{g\ell})^2 + 2g(4\ell)$$

$$v = \sqrt{9g\ell} = 3\sqrt{g\ell}$$

3. Ans (B,C)

$$E \times 4\pi r^2 = \frac{\int_0^r \frac{5}{2\pi} \left(1 - \frac{r}{30}\right) 4\pi r^2 dr}{\epsilon_0}$$

$$\frac{dE_{\text{inside}}}{dr} = 0$$

4. Ans (A,B,C,D)

A

Molecular wt. = $16 M_0$

mass = $2m_0$

$$n_A = \frac{n_0}{8}$$

B

Molecular wt. = M_0

mass = m_0

$n_B = n_0$

(A) K.E/atom = $\frac{f}{2} k.T = \frac{f}{2} k.T$ for both the gases.

$$(B) C_{\text{rms}A} = \sqrt{\frac{3}{16} \frac{RT}{M_0}}, C_{\text{rms}B} = \sqrt{\frac{3RT}{M_0}}$$

$$(C_{\text{rms}})_B = 4(C_{\text{rms}})_A$$

$$(C) (P)_A = \frac{(n_0/8) RT}{V}, (P)_B = \frac{n_0 RT}{V}$$

$$(P)_B = 8(P)_A \quad (D) n_B = 8n_A$$

5. Ans (A,C)

For a cyclic process,

$$\Delta U = 0$$

$$\text{i.e., } \Delta U = \Delta U_1 + \Delta U_2 = 0$$

From relation

$$\Delta Q = \Delta U + \Delta W$$

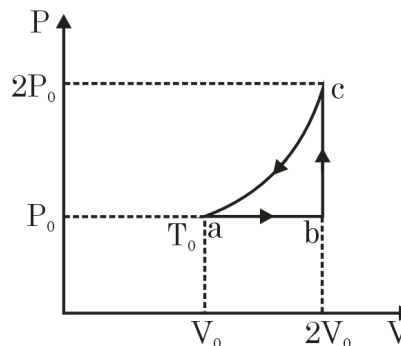
$$\text{As } \Delta U = 0$$

$$\text{Hence, } \Delta Q = \Delta W$$

$$\text{or } \Delta Q - \Delta W = 0$$

6. Ans (A,B,C,D)

One mole of an ideal monatomic gas (initial temperature T_0) is made to go through the cycle abca shown in figure. U denotes the internal energy.



$$\text{For the process ab, } \frac{P_0 V_0}{T_0} = \frac{2P_0 V_0}{T_0}$$

$$T_b = 2T_0$$

$$T_b > T_a \Rightarrow U_b > U_a$$

$$U_b - U_a = C_v \Delta T = \frac{3R}{2} (2T_0 - T_0) = \frac{3RT_0}{2}$$

For the process bc,

$$\frac{P_0 (2V_0)}{2T_0} = \frac{2P_0 (2V_0)}{T_C} \Rightarrow T_C = 4T_0$$

$$T_C > T_0$$

$$U_c - U_b = \frac{3R}{2} (4T_0 - 2T_0) = 3RT_0$$

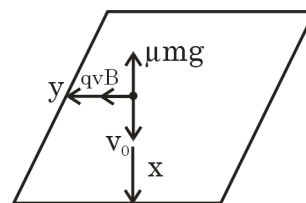
For the process ca,

$$U_c - U_a = \frac{3R}{2} (4T_0 - T_0) = \frac{9RT_0}{2}$$

PART-1 : PHYSICS

SECTION-I (ii)

8. Ans (B,D)



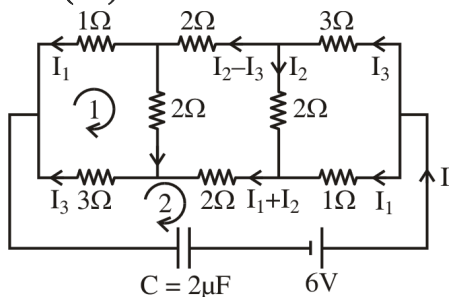
$$(A) qvB = \frac{mv^2}{R} = mv\omega$$

$$\omega = \frac{QB}{m}$$

$$v = v_0 - \mu_g t.$$

$$(B) \omega_r = \Delta k \Rightarrow \ell = \frac{v_0^2}{2\mu_g}$$

10. Ans (C)



$$I_3 - 2I_2 = I_1 \quad \dots(1)$$

$$I_1 + 2I_2 = I_3 \quad \dots(2)$$

$$\therefore I_3 = -2I_2$$

$$I_1 = -4I_2$$

In loop (1)

$$2I_2 + 3I_3 = I_1 \quad \dots(A)$$

Loop (2)

$$6 - 3I_3 - 2(I_1 + I_2) - I_1 = 0$$

$$\Rightarrow 3I_1 + 3I_3 + 2I_2 = 6$$

$$\Rightarrow -12I_2 + 3(-2I_2) + 2I_2 = 6$$

$$\Rightarrow -16I_2 = 6 \Rightarrow I_2 = \frac{-3}{8} \text{ A}$$

Now,

$$I_1 = (-4) \left(\frac{-3}{8} \right) = \frac{3}{2} \text{ A}$$

$$I_3 = (-2) \times \left(\frac{3}{8} \right) = \frac{3}{4} \text{ A}$$

$$I = I_1 + I_3 = \frac{3}{2} + \frac{3}{4} = \frac{9}{4}$$

$$R = \frac{V}{I} = \frac{6}{\frac{9}{4}} = \frac{8}{3} \Omega$$

$$\tau = RC = \frac{8}{3} \times 2 = \frac{16}{3} \mu\text{s}$$

$$I = I_0 e^{-t/\tau} = \frac{9}{4} e^{-\frac{3t}{16}}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (9.00)

$$d\tau = 2i_0 dx B \frac{a}{2} \sin \theta$$

$$\tau = i_0 a \sin \theta \int_0^a B_0 \left(1 + \frac{x}{4} \right) dx$$

$$= 6i_0 a B_0 \sin \theta$$

$$dw = \tau d\theta = 24B_0 I_0 \sin \theta d\theta$$

$$w = 24B_0 I_0 \int_0^{\pi/2} \sin \theta d\theta = 24 \times \frac{1}{8} \times 3 = 9$$

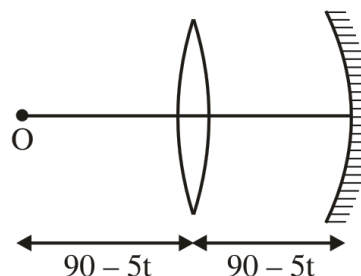
2. Ans (5.00)

$$B_0 a \cdot b = \frac{B_0}{(1+5t)} \int b dx$$

$$\therefore x = a(1+5t)$$

$$v = 5a = 5 \text{ m/s}$$

3. Ans (5.00)



$$90 - 5t = 32$$

$$t = \frac{58}{5}$$

4. Ans (10.00)

$$\tan 37^\circ = \frac{v_y}{v_x} = \frac{v_y}{v_0}$$

$$v_y = a_y t = \frac{qE_y}{m} \times \frac{\ell}{v_0}$$

$$E_y = \frac{v}{d} = \frac{IR}{d} = \frac{\epsilon R}{(R+r)d}$$

$$\Rightarrow \frac{3}{4} v_0 = \frac{16}{91} \times 10^{12} \times \frac{3 \times R}{(R+2) \times 10^{-3}} \times 0.182$$

$$\Rightarrow 2.5R + 5 = 3R$$

$$0.5R = 5$$

$$R = 10 \Omega$$

5. Ans (5.00)

$$v^2 = 2hg$$

$$I = \frac{vB\ell}{R}$$

$$mgv = \frac{v^2 B^2 \ell^2}{R}$$

$$\Rightarrow v = \frac{mgR}{B^2 \ell^2}$$

$$\Rightarrow \frac{m^2 g^2 R^2}{B^4 \ell^4} = 2gh$$

$$h = \frac{m^2 g^2 R^2}{2B^4 \ell^4} = \frac{1 \times 10}{100} \times \frac{1 \times 100 \times 100}{100 \times 2 \times 1} = 5 \text{ m}$$

6. Ans (10.00)

At time t ,

$$mg - BIL = ma \quad \dots(1)$$

$$Q = CvBL$$

$$I = \frac{dQ}{dt} = CBL \frac{dv}{dt} = CBLa \quad \dots(2)$$

From (1) and (2),

$$mg = ma + B^2 L^2 C a$$

$$a = \frac{mg}{m + B^2 L^2 C} = 5$$

$$S = \frac{1}{2}at^2 = \frac{1}{2} \times 5 \times 2^2 = 10 \text{ m}$$

7. Ans (3.00)

$$T_A = T_B = T(\text{say})$$

$$\text{Now } V_A = \frac{nRT}{16P_0} = V$$

$$\Rightarrow V_B = V_C = \frac{nRT}{2P_0} = 8V$$

$$\text{Now in } A \rightarrow C, 16 P_0 V^\gamma = P_0 (8V)^\gamma$$

$$\Rightarrow \gamma = \frac{4}{3} \Rightarrow C_V = \frac{R}{\gamma - 1} = 3R$$

8. Ans (2.00)

$$e = B\ell v = \frac{8}{10} \times 3 \times 5 = 12 \text{ V olt}$$

$$\therefore q = CE (1 - e^{-t/\tau})$$

$$\therefore 24 = 6 \times 12 (1 - e^{-t/\tau}) \Rightarrow e^{-t/\tau} = \frac{2}{3}$$

$$\therefore i = \frac{E}{R} e^{-t/\tau} = \frac{12}{4} \times \frac{2}{3} = 2$$

PART-2 : CHEMISTRY

SECTION-I (i)

2. Ans (A,C,D)

$$\therefore \frac{-d[A]}{dt} = k[A]^{-1}$$

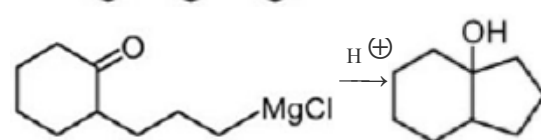
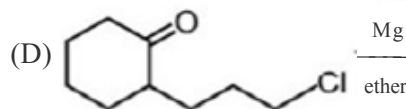
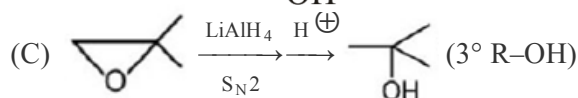
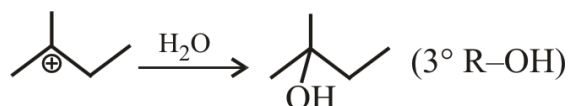
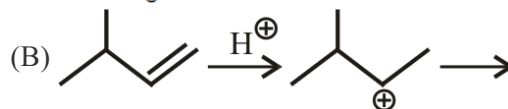
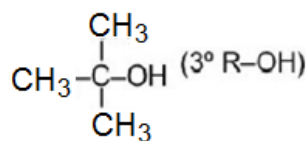
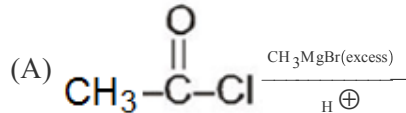
$$\therefore - \int_{a_0}^{a_t} [A] d[A] = \int_0^t k dt$$

$$\therefore \frac{[A]_0^2}{2} - \frac{[A]_t^2}{2} = kt$$

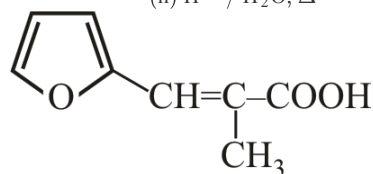
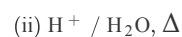
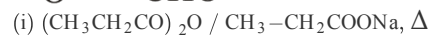
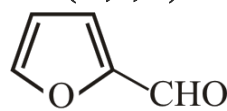
$$\therefore t_{1/2} = \frac{3a_0^2}{8k}$$

Negative order reactions are always complex reaction.

3. Ans (A,B,C,D)



4. Ans (A,C,D)



5. Ans (A,C,D)

$[\text{Pt}(\text{ox})_3]^{2-}$ is optically active due to 3 didentate ligands. $[\text{PdCl}_4]^{2-}$ and $[\text{Pt}(\text{ox})_3]^{2-}$ do not show G.I. as all ligands are same.

Hybridization of Ni^{2+} is dsp^2 in $[\text{Ni}(\text{CN})_4]^{2-}$ and that of Hg^{2+} is sp^3 in $[\text{Hg}(\text{SCN})_4]^{2-}$

Both CN^- and SCN^- ligands are ambidentate. Hence both $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{Hg}(\text{SCN})_4]^{2-}$ show linkage isomerism.

$\text{K}_2[\text{Ni}(\text{CN})_4]$ is diamagnetic but $\text{Co}[\text{Hg}(\text{SCN})_4]$ is not as Co^{+2} is d^7 - paramagnetic.

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (6.00)**

For $z = 1$

$$\therefore \Delta E = 2.55 = 13.6 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$n_1 = 2$$

$$n_2 = 4$$

$$\text{Maximum number of lines} = \frac{4(4-1)}{2} = 6$$

2. **Ans (4.00)**

$$(ii) \Delta H = \Delta U \quad (\Delta n_g = 0)$$

$$\Delta H = \Delta U = q_v$$

(iii) For any irreversible isothermal expansion

$$\Delta S_{\text{universe}} > 0$$

$$\Delta S_{\text{system}} + \Delta S_{\text{surr}} > 0$$

$$\Delta S_{\text{system}} \Rightarrow +ve$$

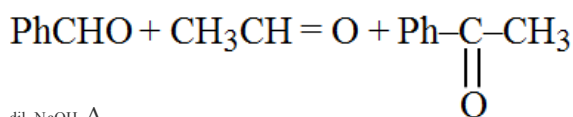
$$\Delta S_{\text{surrounding}} \Rightarrow -ve \therefore |\Delta S_{\text{system}}| > |\Delta S_{\text{surrounding}}|$$

$$\Delta S_{\text{surrounding}}$$

(vii) P (black, solid) $\longrightarrow$ P (white, solid) $\Delta S =$ positive

(viii) For mixing equal values of two ideal gases at constant temperature and pressure in an isolated container, $\Delta S_{\text{system}} > 0$ and $\Delta S_{\text{total}} > 0$

4. **Ans (6.00)**

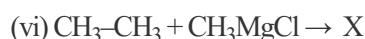
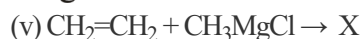
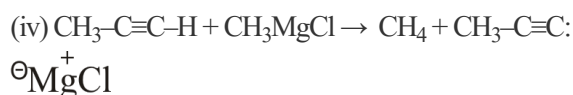
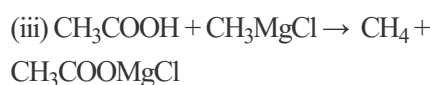


$\xrightarrow{\text{dil. NaOH}, \Delta}$ Total 6 aldol products

6. **Ans (3.00)**

a, c, d

7. **Ans (4.00)**



PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B,D)**

$$t = x^2 + 4x + \alpha^2 - 3\alpha \quad \cot^{-1}t \quad \left(0, \frac{\pi}{2}\right]$$

For function to be onto, $t \geq 0$

$$\text{Hence } x^2 + 4x + \alpha^2 - 3\alpha \geq 0$$

$$D = 16 - 4(\alpha^2 - 3\alpha) = 0 \Rightarrow \alpha = -1, 4$$

$$\text{Hence, } f(x) = \cot^{-1}(x+2)^2$$

$$f'(x) = \frac{-1}{1+(x+2)^2} \cdot 2(x+2) \Rightarrow f'(0) = \frac{-4}{17}$$

$f(x)$ is also many one function as it involves $(x+2)^2$

2. **Ans (A,B,C)**

$$\text{Rf}'(0) = \lim_{h \rightarrow 0} \frac{h + 2h^2 \sin \frac{1}{h} - 0}{h} = 1$$

$$\text{Lf}'(0) = \lim_{h \rightarrow 0} \frac{-h + 2h^2 \sin \frac{1}{-h} - 0}{-h} = 1$$

$$f'(x) = \begin{cases} 1 + 4x \sin \frac{1}{x} - 2 \cos \frac{1}{x} & ; x \neq 0 \\ 1 & ; x = 0 \end{cases}$$

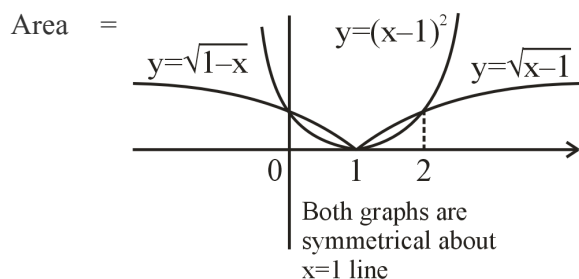
$f'(x)$ is discontinuous at $x = 0$ but $f(x)$ is

differentiable $\forall x \in \mathbb{R}$

$f'(0) > 0$. Hence, $f(x)$ is increasing at $x = 0$

3. Ans (A,B,C,D)

$$y = \sqrt{|x-1|} \text{ and } \sqrt{y} = |x-1|$$



$$2 \int_0^1 [\sqrt{1-x} - (x-1)^2] dx \dots (i)$$

$$\text{or } 2 \int_1^2 [\sqrt{1-x} - (x-1)^2] dx$$

$$\text{Let } x-1 = t$$

$$\text{Area} = 2 \int_0^1 (\sqrt{t} - t^2) dt = 2 \left[\frac{2}{3} - \frac{1}{3} \right] = \frac{2}{3}$$

In (i), substitute $x = t-1$

$$\text{Area} = 2 \int_1^2 [\sqrt{2-t} - (t-1)^2] dt$$

4. Ans (A,B,C)

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x \ln \left(1 + \frac{1}{x} \right) = \lim_{t \rightarrow \infty} \frac{\ln(1+t)}{t}$$

$$= \lim_{t \rightarrow \infty} \frac{\frac{1}{1+t}}{1} = 0$$

$$\lim_{x \rightarrow \infty} f(x) = \lim_{t \rightarrow 0} \frac{\ln(1+t)}{t} = 1$$

$$f(x) = x \ln \left(1 + \frac{1}{x} \right) \Rightarrow f'(x) = \ln \left(\frac{1+x}{x} \right) - \frac{1}{1+x}$$

$$f''(x) = \frac{-1}{x(1+x)^2} < 0 \forall x \in (0, \infty) \Rightarrow f'(x)$$

is $\downarrow$ function

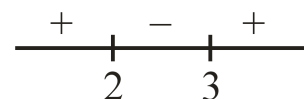
$$x < \infty \Rightarrow f'(x) > f'(\infty)$$

$$f'(x) > \lim_{x \rightarrow \infty} \ln \left(1 + \frac{1}{x} \right) - \frac{1}{1+x} = 0$$

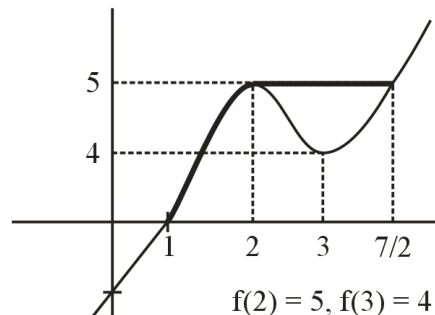
$\Rightarrow f'(x) > 0$ or $f(x)$ is strictly increasing function

5. Ans (A,C,D)

$$f'(x) = 6x^2 - 30x + 36 = 6(x-2)(x-3)$$



$f(x)$ has local maxima at $x=2$ and minima at $x=3$



$$g(x) = \begin{cases} f(x) & ; 1 \leq x \leq 2 \\ 5 & ; 2 < x \leq \frac{7}{2} \\ 12-2x & ; \frac{7}{2} < x \leq 6 \end{cases}$$

$g(x)$ is continuous $\forall x \in [1, 6]$ but non-differentiable at $x = \frac{7}{2}$.

Maximum value of $g(x)$ is 5 which occurs $\forall x \in \left[2, \frac{7}{2} \right]$.

Minimum value of $g(x)$ is zero at $x=1$ and 6.

6. Ans (A,C,D)

Let $P(x, y)$, equation of normal at P

$$\Rightarrow Y - y = \frac{-1}{\left(\frac{dy}{dx} \right)} (X - x)$$

$$A \left(x + y \frac{dy}{dx}, 0 \right)$$

$$B \left(0, y + \frac{x}{\left(\frac{dy}{dx} \right)} \right)$$

$$\frac{BP}{PA} = \frac{1}{2} \Rightarrow y \frac{dy}{dx} = 2x \Rightarrow y^2 = 2x^2 + C$$

$\therefore$ Curve passes through $(0, 4)$

$$\text{So } y^2 = 2x^2 + 16$$

Now check options

PART-3 : MATHEMATICS

SECTION-I (ii)

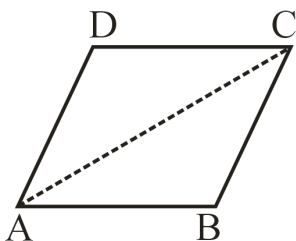
7. Ans (B,D)

$$\frac{4\pi A_{||gm}}{P^2_{||gm}} = \frac{4\pi A_{\Delta}}{P_{\Delta}^2} ; A_{||gm} = 2A_{\Delta}$$

$$\sqrt{2} P_{\Delta} = P_{||gm}$$

$$\sqrt{2} (AB + BC + AC)$$

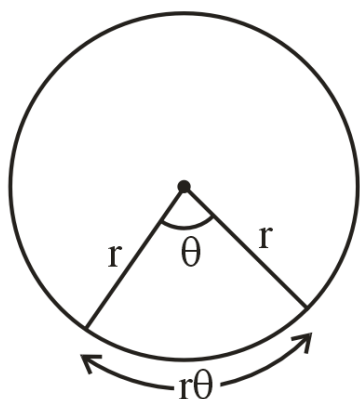
$$= 2(AB + BC)$$



$$\Rightarrow \frac{AB + BC}{AC} = \frac{1}{\sqrt{2} - 1} = \sqrt{2} + 1$$

8. Ans (A,D)

$$IQ = \frac{4\pi A}{P^2} = \frac{4\pi \left(\frac{\theta r^2}{2} \right)}{(2r + r\theta)^2} = \frac{2\pi\theta}{(\theta + 2)^2}$$



$$f(\theta) = \frac{2\pi}{\theta + \frac{4}{\theta} + 4}$$

$$\frac{\theta + \frac{4}{\theta}}{2} \geq \sqrt{\theta \cdot \frac{4}{\theta}} \Rightarrow \theta + \frac{4}{\theta} \geq 4$$

$$\text{or } \theta + \frac{4}{\theta} + 4 \geq 8$$

$f(\theta)$ is maximum when $\theta + \frac{4}{\theta} + 4$ is

minimum which occurs at $\theta = 2$.

9. Ans (A,B,D)

$$\text{Let } \sum_{k=1}^n 3^k (f(x+ky) - f(x-ky)) = S_n \text{ i.e.}$$

$$|S_n| \leq 1$$

$$\Rightarrow -1 \leq S_n \leq 1. \text{ Also } |S_{n-1}| \leq 1 \text{ or } -1 \leq S_{n-1} \leq 1$$

$$\Rightarrow -2 \leq T_n \leq 2 \text{ or } -2 \leq 3^n (f(x+ny) - f(x-ny)) \leq 2$$

$$-\frac{2}{3^n} \leq f(x+ny) - f(x-ny) \leq \frac{2}{3^n} \Rightarrow f(x) \text{ is}$$

constant function

$$\lim_{x \rightarrow 0} \frac{axe^x - b \ln(1+x) + cxe^{-x}}{x^2 \sin x} = 2$$

$$\lim_{x \rightarrow 0} \frac{ax \left(1 + x + \frac{x^2}{2!} + \dots \right) - b \left(x - \frac{x^2}{2} + \frac{x^3}{3!} \dots \right) + cx \left(1 - x + \frac{x^2}{2!} \dots \right)}{x^2 \left(x - \frac{x^3}{3!} + \dots \right)} = 2$$

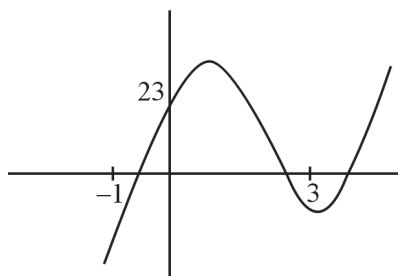
$$\lim_{x \rightarrow 0} \frac{(a-b+c)x + \left(a + \frac{b}{2} - c \right) x^2 + \left(\frac{a}{2} - \frac{b}{3} + \frac{c}{2} \right) x^3 + \dots}{x^3 \left(1 - \frac{x^2}{3!} + \dots \right)} = 2$$

$$a-b+c=0, a + \frac{b}{2} - c=0, \frac{a}{2} - \frac{b}{3} + \frac{c}{2} = 2$$

$$\Rightarrow a=3, b=12, c=9$$

$$\text{Hence, } g(x) = 3x^3 - 15x^2 + 9x$$

$$f(13) = -23 \Rightarrow f(x) = -23$$



$$f(x) = g(x) \Rightarrow 3x^3 - 15x^2 + 9x + 23 = 0$$

$$h(x) = 3x^3 - 15x^2 + 9x + 23$$

$$h(-1) = -4$$

$$h(0) = 23$$

$$h(3) = -4$$

$h(x) = 0$ has 3 distinct roots

10. Ans (A)

$$\text{Let } \sum_{k=1}^n 3^k (f(x+ky) - f(x-ky)) = S_n \quad \text{i.e.}$$

$$|S_n| \leq 1$$

$$\Rightarrow -1 \leq S_n \leq 1. \text{ Also } |S_{n-1}| \leq 1 \text{ or } -1 \leq S_{n-1} \leq 1$$

$$\Rightarrow -2 \leq T_n \leq 2 \text{ or } -2 \leq 3^n (f(x+ny) - f(x-ny)) \leq 2$$

$$-\frac{2}{3^n} \leq f(x+ny) - f(x-y) \leq \frac{2}{3^n} \Rightarrow f(x) \text{ is}$$

constant function

$$\lim_{x \rightarrow 0} \frac{axe^x - b \ln(1+x) + cxe^{-x}}{x^2 \sin x} = 2$$

$$\lim_{x \rightarrow 0} \frac{ax(1+x+\frac{x^2}{2!}+\dots) - b(x-\frac{x^2}{2}+\frac{x^3}{3!}+\dots) + cx(1-x+\frac{x^2}{2!}+\dots)}{x^2(x-\frac{x^3}{3!}+\dots)} = 2$$

$$\lim_{x \rightarrow 0} \frac{(a-b+c)x + (a+\frac{b}{2}-c)x^2 + (\frac{a}{2}-\frac{b}{2}+\frac{c}{2})x^3 + \dots}{x^3(1-\frac{x^2}{3!}+\dots)} = 2$$

$$a-b+c=0, a+\frac{b}{2}-c=0, \frac{a}{2}-\frac{b}{2}+\frac{c}{2}=2$$

$$\Rightarrow a=3, b=12, c=a$$

$$\text{Hence, } g(x) = 3x^3 - 15x^2 + 9x$$

$$\lim_{x \rightarrow 1} \frac{3x^3 - 15x^2 + 9x + 3}{\ln(2-x)}$$

$$\lim_{x \rightarrow 1} \frac{3(x-1)(x^2-4x-1)}{\ln(2-x)}$$

$$= \lim_{h \rightarrow 0} \frac{3h[(1+h)^2 - 4(1+h) - 1]}{\ln(1-h)} = 12$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (0.25)

Chords AB and CD are parallel

Hence, axis is

parallel to

diameter

Slope of diameter

= slope of chord

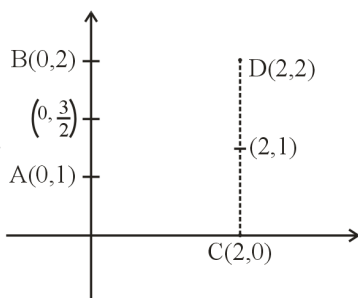
joining mid-

points of

AB and CD

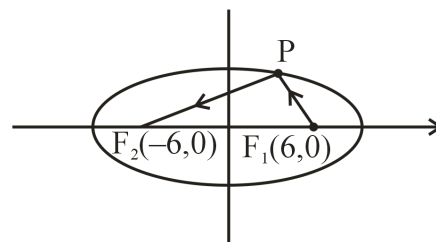
i.e. $(0, \frac{3}{2})$ and $(2, 1)$

$$\text{slope} = \frac{-1}{4}$$



2. Ans (6.50)

$$\frac{x^2}{64} + \frac{y^2}{100} = 1. \text{ Let } x=Y, y=X$$



$$\frac{X^2}{100} + \frac{Y^2}{64} = 1. \text{ Focus } \equiv (\pm 6, 0)$$

Ray F_1P after reflection passes

through focus F_2 (which is B)

$$BP = F_2P = a + ex_p = 10 + \frac{3}{5}(5) = 13$$

3. Ans (0.94 to 0.96)

$$I = \int_0^1 \frac{\sin^{-1} \sqrt{x}}{x(x-1)+1} dx$$

$$I = \int_0^1 \frac{\sin^{-1} \sqrt{1-x}}{(1-x)(-x)+1} dx$$

$$2I = \int_0^1 \frac{\sin^{-1} \sqrt{x} + \sin^{-1} \sqrt{1-x}}{x^2 - x + 1} dx$$

$$\Rightarrow 2I = \int_0^1 \frac{\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}}{x^2 - x + 1} dx$$

$$I = \frac{\pi}{4} \int_0^1 \frac{dx}{(x-\frac{1}{2})^2 + \frac{3}{4}} = \frac{\pi^2}{6\sqrt{3}}$$

4. **Ans (8.00)**

$$(x-16)f(2x) = 16(x-1)f(x) \dots (i)$$

substitute $x = 1$ and $x = 16$, we get $f(2) = f(16) = 0$

$$\text{Hence } f(x) = (x-2)(x-16)g(x)$$

$$\text{from (i), } (x-16)(2x-2)(2x-16)g(2x)$$

$$= 16(x-1)(x-2)(x-16)g(x)$$

$$(x-8)g(2x) = 4(x-2)g(x) \dots (ii)$$

$$\text{Now, } g(4) = g(8) = 0$$

$$\text{Hence, } g(x) = (x-4)(x-8)h(x)$$

$$\text{from (ii), } (x-8)(2x-4)(2x-8)h(2x)$$

$$= 4(x-2)(x-4)(x-8)h(x)$$

$$h(2x) = h(x)$$

$\Rightarrow h(x)$ is a constant function

$$f(x) = \lambda(x-2)(x-16)(x-14)(x-8)$$

$$\text{Given } \lambda = 1 \Rightarrow \frac{f(0)}{128} = 8$$

5. **Ans (1.00)**

$$\text{In } x \in [0, 1]$$

$$\int_0^1 f(x) dx + \int_0^{1/2} f^{-1}(x) dx = \frac{1}{2} \times 1 - 0 \times 0$$

$$\int_0^{1/2} f^{-1}(x) dx = \frac{1}{2} - \int_0^1 \frac{x}{1+x^2} dx$$

$$= \frac{1}{2} - \frac{1}{2} \ln(1+x^2) \Big|_0^1 = \frac{1}{2} - \frac{1}{2} \ln 2$$

$$\text{Area} = 2 \int_0^{1/2} f^{-1}(x) dx = 1 - \ln 2$$

6. **Ans (0.00)**

$$f(x) = \lambda(x-x_1)(x-x_2)(x-x_3)(x-x_4)(x-x_5)$$

$$\ln f(x) = \ln \lambda + \ln(x-x_1) + \dots + \ln(x-x_5)$$

$$\frac{f'(x)}{f(x)} = \frac{1}{x-x_1} + \frac{1}{x-x_2} + \dots + \frac{1}{x-x_5}$$

$$\frac{f''(x)f(x) - (f'(x))^2}{f^2(x)}$$

$$= - \left[\frac{1}{(x-x_1)^2} + \dots + \frac{1}{(x-x_5)^2} \right] < 0$$

Hence, no real roots of

$$f''(x)f(x) - (f'(x))^2 = 0$$

7. **Ans (2.00)**

$$\int (x^9 + x^6 + x^3) (2x^6 + 3x^3 + 6)^{1/3} dx$$

$$= \int (x^8 + x^5 + x^2) (2x^9 + 3x^6 + 6x^3)^{1/3} dx$$

$$\text{let } 2x^9 + 3x^6 + 6x^3 = t^3$$

$$\Rightarrow 18(x^8 + x^5 + x^2)dx = 3t^2 dt$$

$$\int t \frac{3t^2 dt}{18} = \frac{1}{24} (2x^9 + 3x^6 + 6x^3)^{4/3}$$

$$\Rightarrow a = \frac{1}{24}$$

8. **Ans (0.00)**

$$I = \int \frac{\sec(2 + \sec x)}{(2 \sec x + 1)^2} dx = \int \frac{2 \cos x + 1}{(2 + \cos x)^2} dx$$

$$= \int \frac{(2 \cot x \operatorname{cosec} x + \operatorname{cosec}^2 x)}{(2 \operatorname{cosec} x + \cot x)^2} dx$$

$$\text{let } 2 \operatorname{cosec} x + \cot x = t$$

$$I = - \int \frac{dt}{t^2} = \frac{1}{t} + C = \frac{1}{2 \operatorname{cosec} x + \cot x} + C$$

$$I = \frac{\sin x}{2 + \cos x} + C$$

$$\lambda = 1, p = -1$$

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	B	A	C	B	A	B	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	A	A	B	C	A	C	B	C	C
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	100	120	25	1000	9	36	10	69	489	128

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	C	D	B	D	B	B	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	A	C	C	C	D	C	D	B	D
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	3	13	4	1	5	3	2	6	4	6

PART-3 : MATHEMATICS

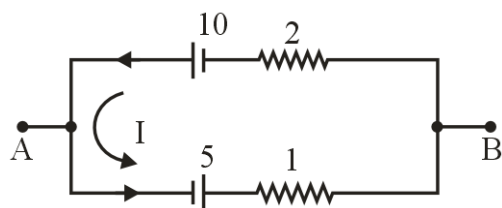
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	B	C	A	C	A	C	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	D	C	B	B	A	B	B	A	C
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	1600	1	10	2	6	784	3	3	3	4

HINT – SHEET
PART-1 : PHYSICS
SECTION-I
1. Ans (D)

Equivalent emf of combination of battery

$$E_{eq} = \frac{\frac{10}{2} - \frac{5}{1}}{\frac{1}{2} + \frac{1}{1}} = 0$$

So current in R will be zero.



$$I = \frac{10 + 5}{2 + 1}$$

$$I = 5A$$

option (D) is incorrect.

2. Ans (C)

 Here $\vec{v} \perp \vec{B}$, B-uniform,

$$v = \sqrt{6^2 + 8^2} = 10 \text{ m/s}$$

So motion of particle will be circular

Its radius is,

$$R = \frac{mv}{qB} = \frac{1 \times 10}{1 \times 2} = 5m$$

time period

$$T = \frac{2\pi m}{qB} = \frac{2 \times (3.14) \times 1}{1 \times 2} = 3.14 \text{ s}$$

3. Ans (B)

$$\phi = \vec{B} \cdot \vec{A} = B_0 e^{-\lambda t} \sqrt{3} \times \frac{a^2}{4}$$

$$\varepsilon = -\frac{d\phi}{dt} = \frac{B_0 \lambda e^{-\lambda t} a^2}{2}$$

$$\& I = \frac{\varepsilon}{3R}$$

4. **Ans (A)**

$$\frac{1}{F} = \frac{1}{F_1} + \frac{1}{F_2} - \frac{1}{F_1 F_2}$$

$$= \frac{1}{10} + \frac{1}{20} - \frac{d}{200}$$

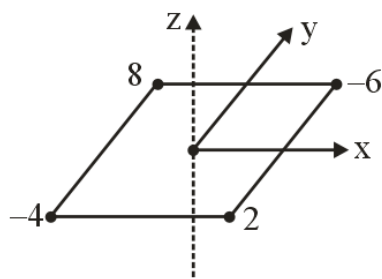
$$\frac{1}{F} = \frac{30-d}{200} \dots (i)$$

if system behaves like concave lens then

F - negative, $30 - d < 0$

$d > 30$

5. **Ans (C)**



Here

$$\Sigma q = 0$$

So potential at each point on z-axis will be zero.

6. **Ans (B)**

$$\text{At } t = 0, \quad x = A, \quad x = A \cos \omega t$$

$$\text{After } t = \frac{T}{6}, \quad x = A \cos \left(\frac{2\pi}{T} \times \frac{T}{6} \right)$$

$$= A \cos 60^\circ$$

$$= \frac{A}{2}$$

$$\therefore \text{Kinetic energy } K = \frac{1}{2} m \omega^2 \left(A^2 - \frac{A^2}{4} \right)$$

$$\text{potential energy } U = \frac{1}{2} m \omega^2 \left(\frac{A}{2} \right)^2$$

$$\frac{K}{U} = \frac{3}{1}$$

7. **Ans (A)**

Equation of parabolic curve is like

$$y = ax^2 \quad a > 0$$

$$\frac{1}{V} = a \cdot T^2$$

$$\therefore VT^2 = \text{constant}$$

$$TV^{1/2} = \text{constant}$$

$$\text{comparing it, } TV^{\gamma-1} = \text{constant}$$

$$\gamma - 1 = \frac{1}{2}$$

$$\gamma = \frac{3}{2}$$

8. **Ans (B)**

Potential difference across A and D

$$= \frac{120}{4} = 30V$$

So potential difference across D and C

$$= 120 - 30 = 90V$$

$$\text{So charge } q_2 = 4 \times 90 = 360 \mu C$$

Here one plate of C_2 and two plates of $4 \mu C$ capacitors connected with one another forms an isolated system. So net charge = 0

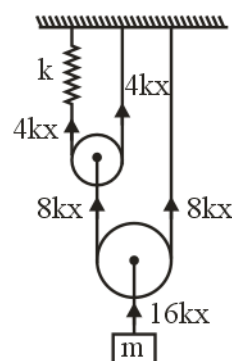
$$-120 + 360 - q_3 = 0$$

$$q_3 = 240 \mu C$$

9. **Ans (C)**

When block is pulled down by x,

spring stretches by 4x. (constrained)



So restoring force on mass m is

$$F_R = -16kx \quad \text{SHM}$$

$$T = 2\pi \sqrt{\frac{m}{16k}} = 2\pi \sqrt{\frac{4}{16 \times 100}}$$

$$T = \frac{\pi}{10}$$

10. **Ans (B)**

If T is temperature of mixture.

Using conservation of energy

$$n_1 \left(\frac{3}{2} kT_1 \right) + n_2 \left(\frac{3}{2} kT_2 \right) = (n_1 + n_2) \times \frac{3}{2} kT$$

$$T = \frac{n_1 T_1 + n_2 T_2}{n_1 + n_2}$$

$$= \frac{(2 \times 300 + 3 \times 200) \times 10^{18}}{(2 + 3) \times 10^{18}}$$

$$T = 240 K$$

11. **Ans (A)**

$$W = P \cdot \Delta V = nR \cdot \Delta T$$

$$Q = nC_P \cdot \Delta T = n \cdot \left(\frac{5R}{2} \right) \Delta T$$

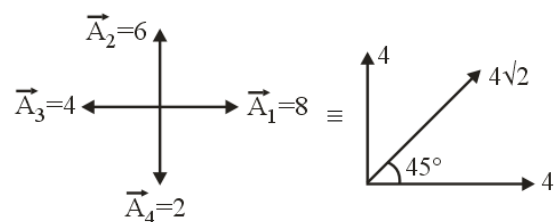
$$\therefore W = \frac{2}{5} Q = \frac{2}{5} \times 100 = 40J$$

12. **Ans (A)**

From resistance codes

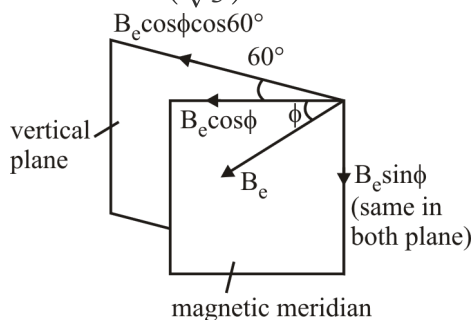
$$R = 14 \times 10^5 \pm 5\%$$

13. **Ans (A)**



14. **Ans (B)**

$$\phi' = \tan^{-1} \left(\frac{2}{\sqrt{3}} \right)$$



$$\tan \phi' = \frac{B_e \sin \phi}{B_e \cos \phi \cos 60^\circ}$$

$$\tan \phi = \tan \phi' \cdot \cos 60^\circ$$

$$= \frac{2}{\sqrt{3}} \times \frac{1}{2} = \frac{1}{\sqrt{3}}$$

$$\phi = 30^\circ$$

15. **Ans (C)**

In case of capacitor $q = CV$

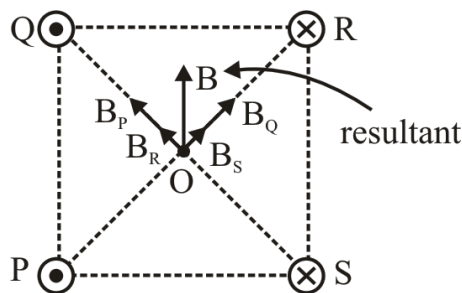
$$\text{current } I = \frac{dq}{dt} = C \left(\frac{dV}{dt} \right)$$

$$\text{Here } \frac{dV}{dt} = \frac{8}{8} = 1V/s \text{ (figure)}$$

So if $C = 2F$

$$I = 2A, \text{ constant}$$

16. **Ans (A)**



17. **Ans (C)**

$$P = 2T^{1/2} = 2 \cdot \left(\frac{PV}{nR} \right)^{1/2}$$

$$PV^{-1} = \text{constant}$$

$$PV^\mu, \mu \rightarrow \text{polytropic index, } \mu = -1$$

$$\therefore C = \frac{R}{(\gamma - 1)} - \frac{R}{(\mu - 1)}$$

$$= \frac{3R}{2} + \frac{R}{2} = 2R$$

18. **Ans (B)**

$$p = \frac{2\pi m}{qB} (v \cos 30^\circ), R = \frac{mv \sin 30^\circ}{qB}$$

$$\frac{R}{p} = \frac{\tan 30}{2\pi} = \frac{1}{2\sqrt{3}\pi}$$

19. **Ans (C)**

Coefficient of performance of heat pump

$$COP = \frac{|Q_H|}{W} \cdot W = 1.2kJ$$

$$= \frac{|Q_H|}{|Q_H| - |Q_C|} = \frac{T_H}{T_H - T_C}$$

$$= \frac{293}{293 - 268} = 11.7$$

$$\text{So } Q_H = COP \times W$$

$$= 11.7 \times 1200 J = 14 kJ$$

20. **Ans (C)**

From conservation of energy

$$\frac{1}{2}mv^2 + \left(\frac{1}{4\pi\epsilon_0} \right) \frac{Qq}{d} = 0 + \left(\frac{1}{4\pi\epsilon_0} \right) \frac{Qq}{r_m}$$

$$r_m = \frac{d}{\left(1 + \frac{2\pi\epsilon_0 mv^2 d}{Qq} \right)}$$

$$= \frac{d}{(1 + 2 \times 2)}$$

$$r_m = \frac{d}{5}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (100)**

For null point $E_0 = k\ell$

Length $AB = L$

$$E_0 = \frac{E}{L} \times (0.3) \quad \{\ell_1 = 30 \text{ cm}\} \dots (i)$$

and for unknown resistance

$$E_0 = \frac{E \cdot 100}{(x + 100)L} \times (0.4) \quad \{\ell_2 = 40 \text{ cm}\} \dots (ii)$$

from equation (i) and (ii)

$$\frac{E}{L} (0.3) = \frac{E \cdot 100}{(x + 100)L} (0.4)$$

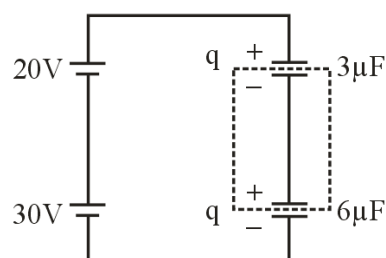
$$\frac{3}{4} = \frac{100}{(x + 100)}$$

$$3x + 300 = 400$$

$$\therefore 3x = 100\Omega$$

2. **Ans (120)**

Initially

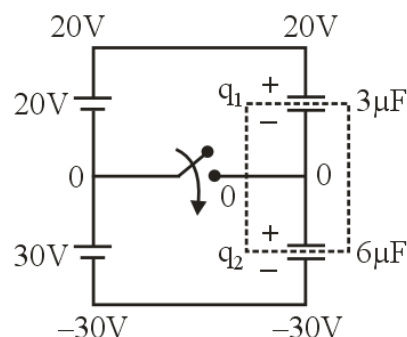


$$C_{eq} = \frac{3 \times 6}{3 + 6} = 2\mu F$$

$$q = C_{eq} V$$

$$q = 100 \mu C$$

when switch S is closed



$$q_1 = 60\mu F$$

$$q_2 = 180 \mu C$$

Charge flown through switch

$$\Delta q = 180 - 60$$

$$\Delta q = 120\mu C$$

3. **Ans (25)**

Using WET

$$(qE_0) x = \frac{1}{2} mv^2 = \frac{1}{2} m(5)^2$$

$$x = \frac{25m}{2qE_0}$$

$$x = \frac{pm}{2qE_0}$$

$$\text{So } p = 25$$

4. **Ans (1000)**

Magnitude of charge flown through loop is

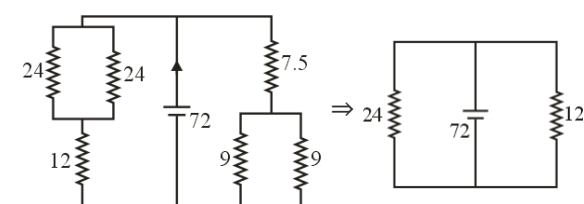
$$\Delta q = \frac{|\Delta \Phi|}{R}$$

$$\Delta q = \frac{2BA}{R} = \frac{2 \times 2 \times 10 \times 10^{-4}}{4}$$

$$\Delta q = 10 \times 10^{-4} C$$

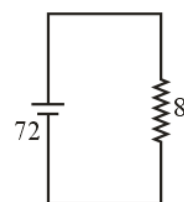
$$\Delta q = 1000 \times 10^{-6} = 1000 \mu F$$

5. **Ans (9)**



$$\Rightarrow R_e = \frac{24 \times 12}{24 + 12} = 8\Omega$$

R_{eq}



$$I = \frac{72}{8} = 9A$$

6. **Ans (36)**

$$\left| \frac{f_0}{f_e} \right| = 6 \Rightarrow f_0 = 6f_e$$

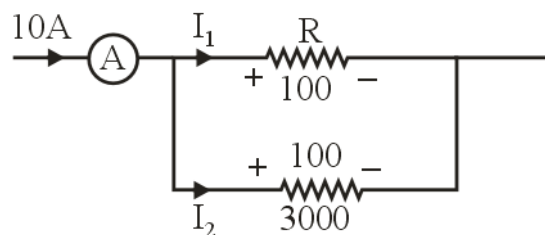
$$L = f_0 + f_e = 42$$

$$7f_e = 42 \Rightarrow f_e = 6 \text{ cm}$$

$$\therefore f_0 = 42 - 6$$

$$f_0 = 36 \text{ cm}$$

7. Ans (10)



$$I = I_1 + I_2$$

$$10 = \frac{100}{R} + \frac{100}{3000}$$

$$R = 10.03\Omega$$

8. Ans (69)

$$\vec{E} = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j} - \frac{\partial V}{\partial z}\hat{k}$$

$$\vec{E} = -\left[(y+z)\hat{i} + (x+z)\hat{j} + (y+x)\hat{k}\right]$$

$$= -10\left[4\hat{i} + 4\hat{j} + 4\hat{k}\right]$$

$$\therefore |\vec{E}| = 40\sqrt{3} = 69.28$$

9. Ans (489)

$$\Delta S_i = \frac{Q}{T} = \frac{mL}{T}$$

$$= \frac{0.4 \times 334 \times 10^3}{273} = 498\text{J/k}$$

10. Ans (128)

As we known

$$\vec{\tau} = \vec{M} \times \vec{B} \dots (i)$$

$$\text{Here } \vec{M} = I \left(\vec{PQ} \times \vec{QR} \right)$$

$$= 8 \left\{ 0.2\hat{k} \times (0.2\hat{i} - 0.2\hat{j}) \right\}$$

$$= 8 \left\{ 0.04\hat{j} + 0.04\hat{i} \right\} = (0.32\hat{i} + 0.32\hat{j}) \text{ Am}^2$$

$$\therefore \vec{\tau} = (0.32\hat{i} + 0.32\hat{j}) \times 0.4\hat{i}$$

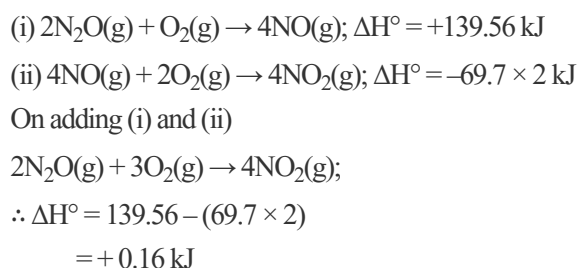
$$\vec{\tau} = -0.128\hat{k} \text{ Nm}$$

$$|\tau| = 128 \times 10^{-3}$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (D)

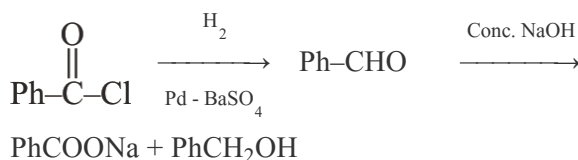


3. Ans (C)

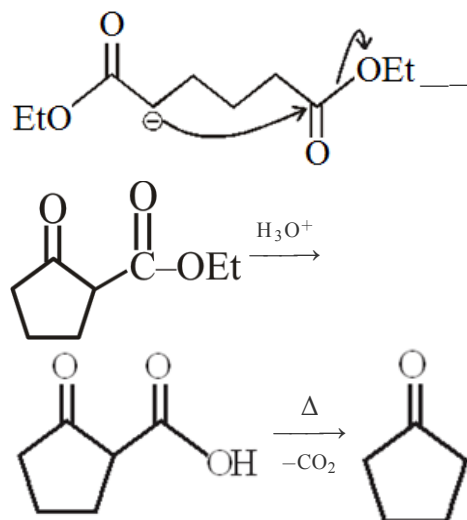
Since 1st step is slow therefore its E_a is high.

Since reaction is exothermic therefore energy of product formed is less than reactant.

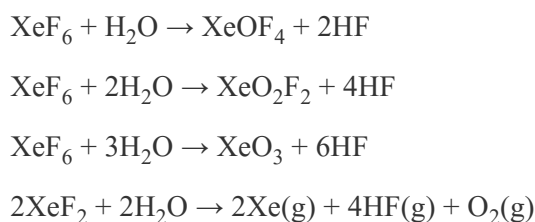
10. Ans (B)



12. Ans (A)



13. Ans (C)



14. **Ans (C)**

(A) Polarising power of cation depends on ionic potential of cation (ϕ).

As ϕ increases polarising power increases

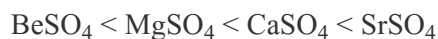
$$\phi = \frac{\text{charge}}{\text{radius}}$$

$\therefore$ charges are same, but size of Be^{2+} is smallest and Ba^{2+} is largest so order should be $\text{Be}^{2+} > \text{Mg}^{2+} > \text{Ca}^{2+} > \text{Ba}^{2+}$

(B) For s-block thermal stability in case of polyatomic anion is decided by covalent nature. As covalent nature increases thermal stability decreases.

BeSO_4 is having maximum covalent nature so least thermal stable

order should be

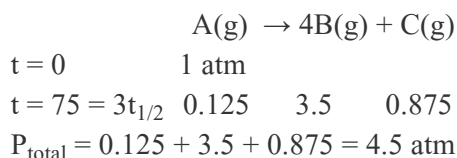


(C) Due to $2p_\pi-2p_\pi$ effective back bonding in BF_3 it lewis acidic nature is least. So given order is correct.

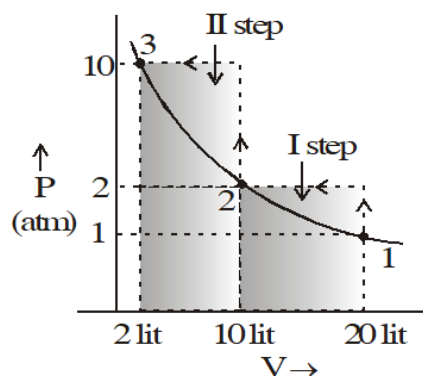
$$\begin{array}{ccc} \text{O}_2 & \text{O}_2^- & \text{O}_2^{2-} \\ \text{(D) B.O.} & \frac{10-6}{2} & \frac{10-7}{2} & \frac{10-8}{2} \\ & = 2.0 & = 1.5 & = 1.0 \end{array}$$

So, order should be $\text{O}_2 > \text{KO}_2 > \text{K}_2\text{O}_2$

18. **Ans (D)**



19. **Ans (B)**



If vol. is 10 lit

$$P_2 = \frac{1 \times 20}{10} = 2 \text{ atm}$$

$$\text{If } V = 2 \text{ lit then } P_3 = \frac{2 \times 10}{2} = 10 \text{ atm}$$

$$W = (20-10) \times 2 + (10-2) \times 10 = 100 \text{ lit-atm}$$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (3)**

$$\text{Wavelength} = \frac{\text{Total length}}{\text{No. of wave}}$$

$$= \frac{1200}{3} = 400 \text{ nm.}$$

$$E = \frac{hc}{\lambda} = \frac{1240}{400} = 3.1 \text{ eV.}$$

2. **Ans (13)**

$$K_f = \frac{R(T_f^0)^2 \times M}{\Delta H_f \times 1000 \times 1000} = 1.86$$

$$\Delta T_f = 2 \times K_f \times m$$

$$2 = 1.86 \times \frac{n_{\text{solute}}}{0.9 \times 18} \times 1000$$

$$n_{\text{solute}} = 0.0174$$

When solution is placed at temperature below its freezing point ($T = 271 \text{ K}$), some of the water freezed out.

$\therefore$ For resulting solution

$$P = P^\circ, X_{\text{solvent}}$$

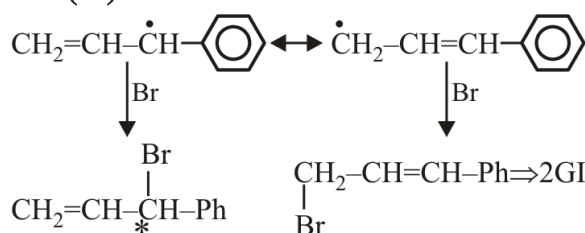
$$700 = 760 \times \frac{n_{\text{H}_2\text{O}}}{n_{\text{H}_2\text{O}} + 0.0174}$$

$$\therefore n_{\text{H}_2\text{O}} = 0.2 \text{ mole}$$

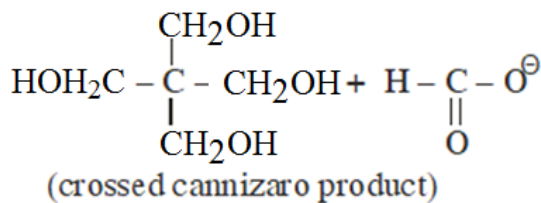
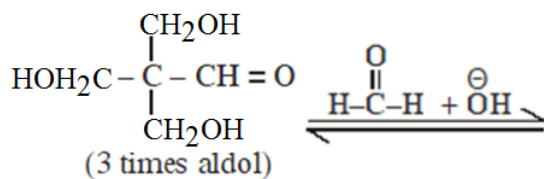
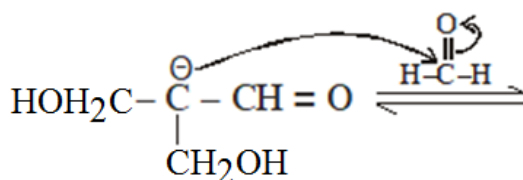
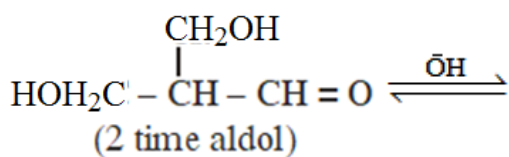
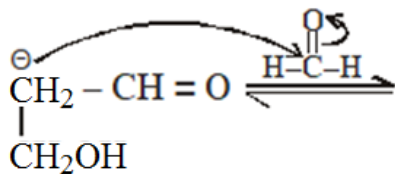
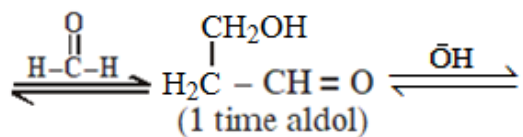
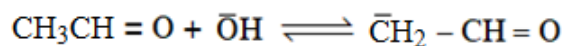
$$\therefore \text{mole of } \text{H}_2\text{O(s)} \text{ formed} = 0.9 - 0.2 = 0.7$$

$$\therefore \text{wt. of } \text{H}_2\text{O(s)} \text{ formed} = 0.7 \times 18 = 12.6 \text{ gm.}$$

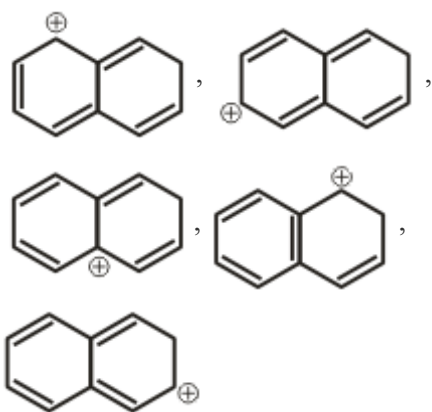
3. **Ans (4)**



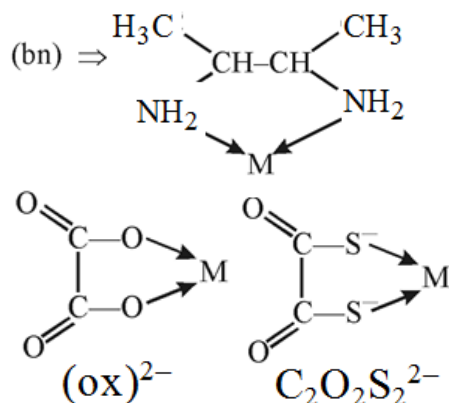
4. Ans (1)



5. Ans (5)



6. Ans (3)



7. Ans (2)

Refer Notes

10. Ans (6)

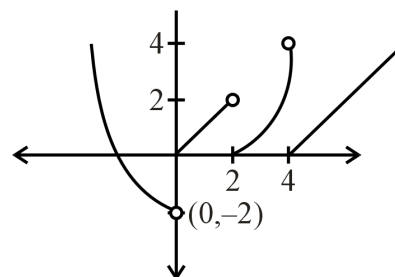
$$\Delta S = nC_{v,m} \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1} = 6.3$$

$$\frac{T_2}{T_1} = \frac{P_2 V_2}{P_1 V_1} ; (C_{v,m} = 5/2 \times R)$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (B)



2. Ans (C)

Check at $x = 0, 1, 2$

at $x = 0$

LHL $f(x) = 1$

RHL and $f(0) = 1$

at $x = 1$

LHL $f(x) = 2$

RHL and $f(1) = 2$

at $x = 2$

LHL $f(x) = 4$

$$\text{RHL and } f(2) = \lim_{x \rightarrow 2} \frac{x^2 - 1}{x - 1} = 3$$

$f(x)$ is discontin. at $x = 2$

3. **Ans (B)**

$$y = x + e^x$$

$$\text{at } x = 1, y = 1 + e$$

$$y' = 1 + e^x; y'' = e^x$$

$$\frac{d^2x}{dy^2} = -\left(\frac{dy}{dx}\right)^{-3} \left(\frac{d^2y}{dx^2}\right)$$

$$= -(1 + e)^{-3}(e)$$

4. **Ans (C)**

$$f''(x) - f'(x) \leq 0$$

$$e^{-x}f'' - e^{-x}f' \leq 0$$

$$\frac{d}{dx}(e^{-x}f'(x)) \leq 0$$

$e^{-x}f'(x)$ is decreasing function

$$e^{-x}f'(x) \leq e^{-0}f'(0) = 0$$

$$f'(x) \leq 0$$

$$\text{but } f''(x) \leq f'(x)$$

$$\text{and } f''(x) \leq 0$$

$$\text{So } f'(x) = 0$$

$$f(x) = \text{const.}$$

$$f(x) = 5$$

5. **Ans (A)**

If $f(x)$ is cont. at $x = 0$

$$\Rightarrow P = r$$

$$f'(x) = \begin{cases} 3 + \frac{2}{\sqrt{1-x^2}} & ; x \leq 0 \\ 4x^3 - 3x^2 + 2x - q & ; x > 0 \end{cases}$$

$$\text{LHD} = \text{RHD at } x = 0$$

$$-q = 5$$

$$q = -5$$

6. **Ans (C)**

$$f(x) = \log_{1/3}(x^2 - 5x + 6)$$

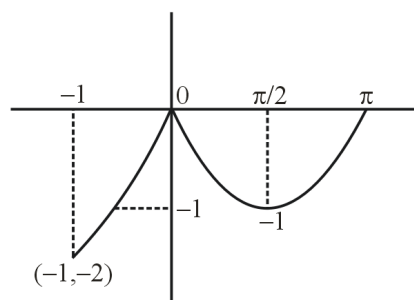
$f(x)$ is inc. so

$$f(x) \geq 0$$

$$\frac{1}{\ln\left(\frac{1}{3}\right)} \frac{1}{(x^2 - 5x + 6)} (2x - 5) \geq 0$$

Now solve

7. **Ans (A)**



8. **Ans (C)**

$$x + y = x^y$$

$$\text{at } x\text{-axis } y = 0, x = 1$$

Point (1, 0)

$$1 + \frac{dy}{dx} = y \cdot x^{y-1} + x^y \ln x \frac{dy}{dx}$$

$$\text{Put (1, 0)} \quad \frac{dy}{dx} = -1$$

$$\text{equation of normal } y - 0 = 1(x - 1) \Rightarrow y = x - 1$$

9. **Ans (B)**

$$I = \int \frac{x^4 + x^3 - 1}{x^2 \sqrt{x^4 + 1 + 2x^3}} dx$$

$$I = \int \frac{x^4 + x^3 - 1}{x^3 \left(x^2 + \frac{1}{x^2} + 2x\right)^{1/2}} dx$$

$$\text{Put } x^2 + \frac{1}{x^2} + 2x = t$$

$$x - \frac{1}{x^3} + 1 = \frac{dt}{2}$$

$$I = \frac{1}{2} \int \frac{dt}{\sqrt{t}}$$

10. **Ans (B)**

$$I = \int_{\pi/4}^{\pi/2} [|\sin x| + |\cos x| + [\operatorname{cosec} x]] dx$$

$$I = \int_{\pi/4}^{\pi/2} ([|\sin x| + |\cos x|] + [\operatorname{cosec} x]) dx$$

$$I = \int_{\pi/4}^{\pi/2} \left[\sqrt{2} \sin\left(x + \frac{\pi}{4}\right) \right] dx + \int_{\pi/4}^{\pi/2} 1 dx$$

because range of $\sqrt{2} \sin\left(x + \frac{\pi}{4}\right)$ in

$$x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right) \text{ is } [1, \sqrt{2}]$$

$$\text{so } \left[\sqrt{2} \sin\left(x + \frac{\pi}{4}\right) \right] = 1$$

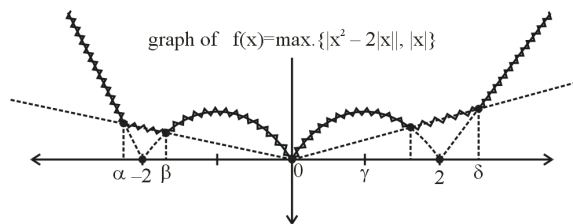
11. **Ans (C)**

$$\sin\left(\tan^{-1}\left(\frac{4}{3}\right)\right) = \sin^{-1}\left(\frac{4}{5}\right) = \frac{4}{5}$$

$$\tan\left(\cos^{-1}\left(\frac{4}{5}\right)\right) = \tan\left(\tan^{-1}\left(\frac{3}{4}\right)\right) = \frac{3}{4}$$

$$\cos\left(\sin^{-1}\left(\frac{3}{4}\right)\right) = \cos\left(\cos^{-1}\left(\frac{\sqrt{7}}{4}\right)\right) = \frac{\sqrt{7}}{4}$$

12. Ans (D)



$f(x)$ is non-differentiable at $x = \alpha, \beta, 0, \gamma, \delta$

13. Ans (C)

Point of intersection of curves

$$y = |x^2 - 1| \text{ and } y = \sqrt{7 - x^2}$$

is $(\pm\sqrt{3}, 2)$

$$\text{for } y = x^2 - 1; \left. \frac{dy}{dx} \right|_{\sqrt{3}, 2} = m_1 = 2\sqrt{3}$$

$$\text{for } y = \sqrt{7 - x^2}; \left. \frac{dy}{dx} \right|_{\sqrt{3}, 2} = m_2 = \frac{-\sqrt{3}}{2}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{5\sqrt{3}}{4} \right|$$

14. Ans (B)

$$\text{Area} = \int_{2/3}^1 (f(x) - g(x)) dx$$

15. Ans (B)

$$\text{Put } x - \frac{\pi}{4} = t \Rightarrow dx = dt$$

$$\begin{aligned} I &= \int_{-\pi/2}^{\pi/2} \frac{1 + \sin t}{1 - \cos\left(\left|\frac{\pi}{4} + t\right| + \frac{\pi}{4}\right)} dt \\ &= \int_{-\pi/2}^{-\pi/4} \frac{1 + \sin t}{1 - \cos t} dt + \int_{-\pi/4}^{\pi/2} \frac{1 + \sin t}{1 + \sin t} dt \\ &= \int_{-\pi/2}^{-\pi/4} \frac{dt}{1 - \cos t} + \int_{-\pi/4}^{\pi/4} \frac{\sin t}{1 - \cos t} dt + \frac{3\pi}{4} \\ &= \left[-\operatorname{cosec} t - \cot t \right]_{-\pi/2}^{-\pi/4} + \ln\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right) + \frac{3\pi}{4} \\ &= \ln\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right) + \frac{3\pi}{4} + \sqrt{2} \end{aligned}$$

16. Ans (A)

$$a \sin \theta + b \cos \theta = 1 \quad \dots(1)$$

$$a \cos \theta - b \sin \theta = \sin \theta \cos \theta \quad \dots(2)$$

squaring and adding

$$a^2 + b^2 = 1 + \sin^2 \theta \cos^2 \theta$$

$$a^2 + b^2 = 1 + \cos^2 \theta - \cos^4 \theta \quad \dots(3)$$

$$(1) \times \cos \theta - (2) \times \sin \theta$$

$$a \sin \theta \cos \theta + b \cos^2 \theta - a \cos \theta \sin \theta + b \sin^2 \theta$$

$$= \cos \theta - \sin^2 \theta \cos \theta$$

$$b = \cos^3 \theta \Rightarrow \cos \theta = b^{1/3}$$

Put in (3)

$$a^2 + b^2 = 1 + b^{2/3} - b^{4/3}$$

$$k = 1$$

$$\beta + \int_0^1 x \cdot 2x e^{-x^2} dx = \int_0^1 e^{-x^2} dx$$

$$\beta + \left[-x e^{-x^2} \right]_0^1 - \int_0^1 e^{-x^2} dx = \int_0^1 e^{-x^2} dx$$

$$\beta = \frac{1}{e}$$

17. Ans (B)

$$I = \int \frac{x^6(x+1)}{\sqrt{5x^{10} + 6x^9 + x^4}} dx$$

$$I = \int \frac{x^5 + x^4}{\sqrt{5x^6 + 6x^5 + 1}} dx$$

$$\text{Put } 5x^6 + 6x^5 + 1 = t$$

$$(x^5 + x^4) dx = dt \frac{1}{30}$$

$$I = \frac{1}{30} \int \frac{dt}{\sqrt{t}}$$

18. Ans (B)

$$f'(x) = f(x) + A \quad \dots(1)$$

$$\frac{dy}{dx} = y + A$$

$$\text{let } A = \int_0^1 f(x) dx$$

$$\ln(y + A) = x + C$$

$$f(0) = 1$$

$$\ln(1 + A) = 0 + C \Rightarrow C = \ln(1 + A)$$

$$\ln(y + A) = x + \ln(1 + A)$$

$$\frac{y + A}{1 + A} = e^x \Rightarrow y + A = (1 + A) e^x \quad \dots(2)$$

$$y = (1 + A) e^x - A$$

$$A = \int_0^1 ((1 + A) e^x - A) dx$$

$$A = (1 + A) \int_0^1 e^x dx - A \int_0^1 dx$$

$$A = (1 + A) (e - 1) - A$$

now put value of A in equation (2)

19. Ans (A)

$$y^2 = 4x ; a = 1$$

$$\text{let } P(at_1^2, 2at_1) \equiv (t_1^2, 2t_1)$$

$$Q(at_2^2, 2at_2) \equiv (t_2^2, 2t_2)$$

intersection of tangents at

P & Q is T(at₁t₂, a(t₁ + t₂))

$$(t_1 t_2, t_1 + t_2)$$

eqⁿ of normal at P

$$y = -tx + 2t + t^3$$

passing through (9, 6)

$$6 = -9t + 2t + t^3$$

$$t^3 - 7t - 6 = 0$$

$$(t + 1)(t^2 - t - 6) = 0$$

$$(t + 1)(t - 3)(t + 2) = 0$$

$$t = -2, -1, 3$$

$$P(4, -4)$$

$$Q(1, -2)$$

$$R(9, 6)$$

$$T(2, -3)$$

20. Ans (C)

$$y = mx + c$$

⊥ distance from origin = 1

$$\frac{|c|}{\sqrt{m^2 + 1}} = 1$$

$$c^2 = m^2 + 1$$

$$(y - mx)^2 = m^2 + 1$$

$$\left(y - x \frac{dy}{dx}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$$

PART-3 : MATHEMATICS

SECTION-II

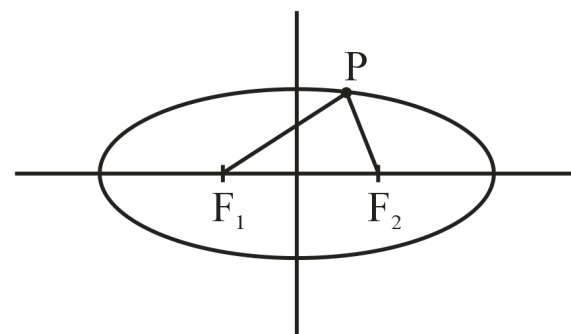
1. Ans (1600)

eccentricity of $xy = c^2$ is $\sqrt{2}$

eccentricity of $x^2 - y^2 = a^2$ is $\sqrt{2}$

$$e_1 = \sqrt{2}, e_2 = \sqrt{2}$$

2. Ans (1)



$$PF_1 + PF_2 = 2a$$

$$F_1 F_2 = 2ae$$

$$2ae = \sqrt{(24 - 5)^2 + (12 - 7)^2}$$

$$= \sqrt{361 + 25} = \sqrt{386}$$

$$2a = \sqrt{144 + 25} + \sqrt{576 + 49} = 38$$

3. Ans (10.00)

$$\int \frac{(x^2 + 1)^2 - 2x^2}{(x^6 + 1)} dx$$

$$\Rightarrow \int \frac{(x^2 + 1)}{x^4 - x^2 + 1} dx - \int \frac{2x^2}{x^6 + 1} dx$$

$$\Rightarrow \int \frac{1 + \frac{1}{x^2}}{(x^2 + \frac{1}{x^2} - 1)} dx - \frac{2}{3} \int \frac{dt}{t^2 + 1} \quad (x^3 = t)$$

$$= \tan^{-1} \left(x - \frac{1}{x} \right) - \frac{2}{3} \tan^{-1}(x^3) + c$$

$$\therefore A + B = \frac{1}{3}$$

4. **Ans (2)**

$f(x)$ is an odd function

$g(x)$ is $f^{-1}(x)$

$$\text{Area} = \int_0^{2\pi} f^{-1}(y) dy$$

put $y = f(x)$

$$\text{Area} = \int_0^{2\pi} x f'(x) dx$$

$$= [xf(x)]_0^{2\pi} - \int_0^{2\pi} (x + \sin x) dx$$

$$= 2\pi^2$$

5. **Ans (6.00)**

$$\int \frac{1}{(1 + \sqrt{x})^8} dx$$

$$x = t^2$$

$$\int \frac{2tdt}{(1+t)^8}$$

$$= 2 \int \frac{t+1-1}{(1+t)^8}$$

$$= 2 \int \frac{1}{(1+t)^7} - 2 \int \frac{1}{(1+t)^8}$$

$$= -\frac{1}{3(1+t)^6} + \frac{2}{7(1+t)^7} + C$$

$$= -\frac{1}{3(1+\sqrt{x})^6} + \frac{2}{7(1+\sqrt{x})^7}$$

$$a = 6, b = 7$$

6. **Ans (784)**

Let the mid point of chord AB is (h, k) the eqⁿ of AB is $T = S_1$

$$49hx + 16ky - 784 = 49h^2 + 16k^2 - 784$$

$$49hx + 16ky = 49h^2 + 16k^2 \quad \dots (1)$$

Any point on the circle $x^2 + y^2 = 100$ is

$$P(10\cos\theta, 10\sin\theta)$$

eqⁿ of chord of contact AB is

$$49x(10\cos\theta) + 16y(10\sin\theta) = 784$$

(1) & (2) are same line AB

$$\frac{49h}{490\cos\theta} = \frac{16k}{160k\sin\theta} = \frac{49h^2 + 16k^2}{784}$$

$$\cos\theta = \frac{784k}{10(49h^2 + 16k^2)}$$

$$\sin\theta = \frac{784h}{10(49h^2 + 16k^2)}$$

$$\cos^2\theta + \sin^2\theta = 1$$

put in this eqⁿ and get Ans.

7. **Ans (3)**

$$\text{Slope of tangent to } \frac{dy}{dx} = 2x$$

$$\text{slope} = 6$$

$$\text{eqⁿ of tangent } y - 13 = 6(x - 3)$$

$$6x - y - 5 = 0$$

it passes through (a, b)

$$6a - b - 5 = 0 \quad \dots (1)$$

slope of tangent to circle at (a, b)

$$2x + 2y \frac{dy}{dx} - 4 + 2 \frac{dy}{dx} \geq 0$$

$$\frac{dy}{dx} = \frac{2-a}{b+1} = 6 \Rightarrow 2-a = 6b+6$$

$$6b + a + 4 = 0 \quad \dots (2)$$

solve (1) & (2)

$$a = \frac{26}{37}, b = \frac{-29}{37}$$

8. **Ans (3)**

$$I = \int_0^3 \frac{f(x)}{x^2 - 3x + 1} dx$$

apply king property

$$I = \int_0^3 \frac{f(3-x)}{x^2 - 3x + 1} dx$$

add both

$$2I = \int_0^3 \frac{f(x) + f(3-x)}{x^2 - 3x + 1} dx$$

$$2I = \int_0^3 dx = [x]_0^3$$

$$I = \frac{3}{2}$$

9. **Ans (3)**

$$\text{for } x \in \left[1, \frac{\pi}{2}\right]$$

$$f(x) = \sqrt{\sin x}$$

$$\text{Area} = \int_1^{\pi/2} \sqrt{\sin x} dx$$

$$= \sqrt{\sin x} \left(\frac{\pi}{2} - 1 \right)$$

10. **Ans (4)**

Area = area inside square form at $x + y = \pm 1$,

$$x - y = \pm 1$$

$$\text{and outside the circle } x^2 + y^2 = \frac{1}{2}$$

$$\text{Area} = 4 \times \frac{1}{2} \times 1 \times 1 - \pi \left(\frac{1}{\sqrt{2}} \right)^2$$

$$= 2 - \frac{\pi}{2}$$